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Authors for correspondence:
Julián García
e-mail: julian.garcia@monash.edu
Arne Traulsen
e-mail: traulsen@evolbio.mpg.de

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Evolution of coordinated punishment to enforce cooperation from an unbiased strategy space

Julián García[1] and Arne Traulsen[2]

[1]Faculty of Information Technology, Monash University, Melbourne, Australia
[2]Department of Evolutionary Theory, Max Planck Institute for Evolutionary Biology, 24306 Plön, Germany

AT, 0000-0002-0669-5267

The emergence and maintenance of punishment to protect the commons remains an open puzzle in social and biological sciences. Even in societies where pro-social punishing is common, some individuals seek to cheat the system if they see a chance to do so—and public goods are often maintained in spite of cheaters who do not contribute. We present a model accounting for all possible strategies in a public goods game with punishment. While most models of punishment restrict the set of possible behaviours, excluding seemingly paradoxical anti-social strategies from the start, we show that these strategies can play an important role in explaining large-scale cooperation as observed in human societies. We find that coordinated punishment can emerge from individual interactions, but the stability of the associated institutions is limited owing to anti-social and opportunistic behaviour. In particular, coordinated anti-social punishment can undermine cooperation if individuals cannot condition their behaviour on the existence of institutions that punish. Only when we allow for observability and conditional behaviours do anti-social strategies no longer threaten cooperation. This is due to a stable coexistence of a minority supporting pro-social institutions and those who only cooperate if such institutions are in place. This minority of supporters is enough to guarantee substantial cooperation under a wide range of conditions. Our findings resonate with the empirical observation that public goods are resilient to opportunistic cheaters in large groups of unrelated individuals. They also highlight the importance of letting evolution, and not modellers, decide which strategies matter.

1. Introduction

Most modern societies have put in place institutions that support and promote collective action. Understanding the origin of these institutions is an important challenge across biological and social sciences [1]. The outstanding capacity of humans to engage in large-scale cooperation often relies on these institutionalized enforcement mechanisms [2]. Centralized institutions for cooperation also have experimental and empirical support [3,4], but explaining how these institutions arise from individual incentives is an open problem [5]. Here, we propose that these institutions play a role in enabling cooperation, not only by implementing punishment against free-riders [6–8], but also by means of their visibility, which enables agents to condition their actions on whether these institutions are present or not.

Punishment provides a possible solution to the problem of collective action [9–14]. The vast majority of theoretical and experimental work focuses on pro-social peer punishment, exerted by peers and individually directed towards those who do not cooperate [15–20]. Anti-social peer punishment is instead directed towards those who do contribute to the public good [21]. Experiments and models have shown that anti-social punishment can diminish the effectiveness of punishment in promoting cooperation [22–25]. In many instances,

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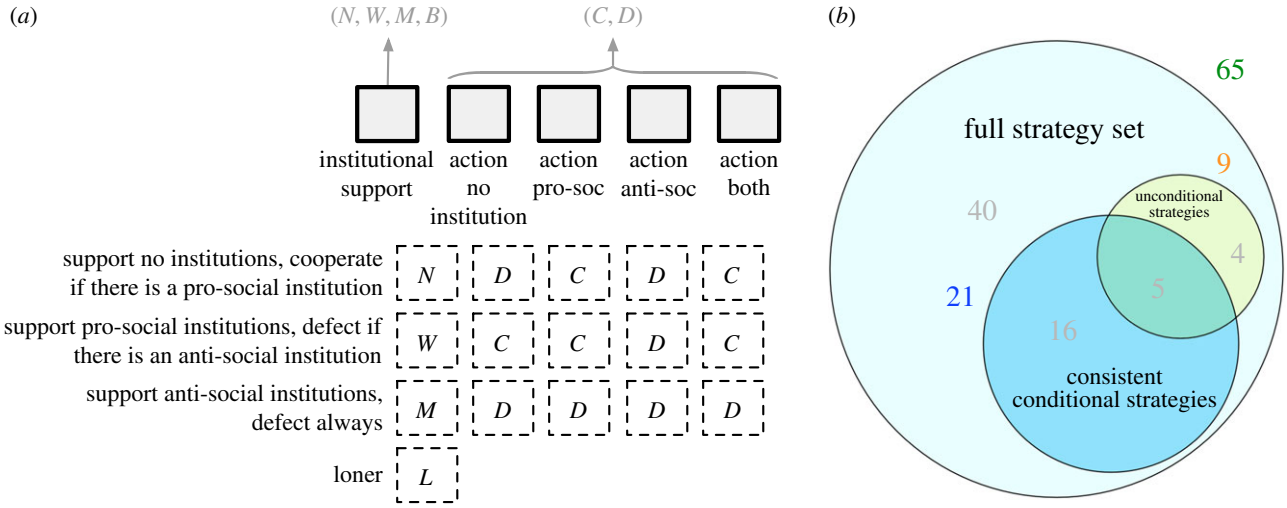

Figure 1. Description of the strategy sets. (*a*) The 64 strategies taking part in the game are characterized by their contribution to the punishment pool (none (*N*), pro-social (*W*), anti-social (*M*) or both (*B*)) and their action for each institutional set-up. In addition, in an optional game, we have the loner strategy which neither punishes nor gets punished. (*b*) The 65 strategies can be reduced to 21 by assuming consistent actions—players do not support pools that punish them. Alternatively, one can focus on nine unconditional strategies, where actions are not affected by the punishment set-up. Iterated removal of dominated strategies results in five strategies that are also consistent. (Online version in colour.)

however, punishment is not individual, but a coordinated action of many individuals [26]. An extreme form of such coordination is the kind of pool punishment that emerged as the typical way of punishment in modern human societies: individuals commit an investment into a pool to pay for the punishment of those who do not comply with a social norm [6,27–29]. Here, we show that pro-social punishment can withstand the presence of cheaters and anti-social behaviour, but this outcome only emerges when considering *all* possible strategies in a public goods game. This result highlights the importance of avoiding artificial restrictions in the strategy set of evolutionary models.

We study the evolutionary dynamics of coordinated anti-social punishment and ask whether the associated coordinated punishment can emerge, potentially undermining cooperation. Our model allows for the evolutionary competition between anti-social and pro-social punishment. We consider two main scenarios. (i) Observable institutions allow individuals to condition their actions on the existence of punishment institutions. (ii) If institutions cannot be observed, individuals are unable to condition their behaviour. When punishment is not observable, anti-social punishment triggers a collapse of the public good. If institutions are observable, cooperation can be established and stabilized by pro-social punishment, even in the face of anti-social behaviour.

2. The model

2.1. A game of cooperation with institutional punishment

Our model follows Sigmund *et al.* [6] in the basic set-up of an optional public goods game between n players with three stages. (i) The first stage is institutional commitment, in which players may commit funds to an institution that will later punish free-riders or cooperators. (ii) The second stage is the public goods game, in which individuals may decide whether to contribute or not to a public good. (iii) The third stage is punishment, in which players are fined in accordance to the institutions in place.

In the institutional commitment stage (i), participants choose what kind of institution they want to support. They can support pro-social punishment directed to defectors, or anti-social punishment of cooperators, both, or none. Funding an institution costs a fixed amount γ if the punishment takes place. An institution is established—and therefore costly—only if there are at least k players contributing to it.

(ii) During the public goods stage, players use the information about the institutions in place, choosing whether they contribute an amount $c > 0$. Contributions are multiplied by a factor $r > 1$ and distributed among the $n - 1$ other players [30].

(iii) During the punishment stage, agents are fined according to the institutions in place, and the amount of players supporting the corresponding institutions. Non-contributors are punished by an amount β multiplied by the number of supporters of the pro-social institution. Contributors are punished by an amount β multiplied by the number of supporters of the anti-social pool.

Since the game is optional, we also let agents opt out of the game altogether. Those that do not take part in the game obtain a loner pay-off $\sigma > 0$, regardless of the decisions of others [31]. An optional game, therefore, includes the 'loner' strategy, whereas a non-optional game precludes it.

When agents can decide what to do depending on the existence of punishment institutions, a large strategy set emerges, as follows. In the first stage of the game, an individual decides their institutional support; with options for supporting no institution (*N*), only a pro-social institution (*W*), only an anti-social institution (*M*) or both institutions (*B*)—thus institutional support entails four possibilities. In addition, agents also decide whether to contribute or not to the public good, contingent on the institutional arrangement in place; i.e. cooperate or defect given there is no institution, cooperate or defect when there is a pro-social institution only, what to do when there is only an anti-social institution, and what to do when both institutions are in place—thus, we have 2^4 possibilities. This yields in total $4 \times 2^4 = 64$ strategies. If we further make the game optional and include the loner option, we obtain a total of 65 strategies (figure 1*a*).

In the analysis that follows we will focus on different subsets of this large strategy set. These subsets will also imply different assumptions in the game. In particular, we study a set of nine non-conditional strategies, in which cooperation or defection does not depend on institutional arrangements—this is equivalent to institutions that cannot be observed. We also study the set of all 65 strategies, whose analysis will be shown equivalent to that arising from the set of 21 consistent strategies. A consistent strategy is such that a player will not contribute to an institution that would punish her actions (figure 1*b*).

2.2. Evolutionary dynamics

The game described above determines the pay-off of each player in the population. We calculate the average pay-off across all possible configurations of groups given the current numbers of each type in the population, such that all players using the same strategy have the same pay-off π. This pay-off determines how many players will adopt the corresponding strategy, as successful strategies spread in a finite population in proportion to their relative fitness. We consider a Moran process, where a single individual chooses a new strategy in each time step with probability proportional to fitness f. We assume an exponential pay-off to fitness mapping [32,33], such that fitness is given by $f = \exp[+\omega\pi]$, where ω is the intensity of selection; see electronic supplementary material for details. In addition, there is a small probability μ that an individual switches to a new random type. In our simulations, we focus on the case of population size $N = 50$, mutation rate $\mu = 0.001$, and intensity of selection $\omega = 10$.

3. Results

We study which strategies are favoured by an evolutionary process. Under small mutation rates, the dynamics of the evolutionary process is confined to edges between two strategies [34,35]. Therefore, it is instructive to first compare pay-offs between any two strategies. A full overview of the $(1/2) \times 65 \times 64 = 2080$ strategy pairs is possible, but hard to grasp. We thus start by reducing the size of this large strategy set, making further assumptions on the nature of possible strategies.

3.1. Non-observable institutions

In the simplest case, individuals cannot condition their actions on the existence of a punishment institution [6]. This is equivalent to punishment institutions that cannot be observed. This case leads to nine strategies, as follows. Individuals have four options to support institutions multiplied by two possible actions in the public goods game. In addition, individuals can choose to abstain from the game. Out of these nine strategies, four are dominated by others. However, instead of neglecting these strategies from the beginning, we include them in our computational model and let evolution decide whether they play any role. Pro-social institutions can promote temporary cooperation, even when fines are exclusively directed towards defectors and not used to stabilize punishment [6]. But this kind of model assumes that anti-social institutions are excluded. When allowing for anti-social institutions, cooperation can not only be undermined by defectors not supporting any institution but also by defectors that in addition set up an anti-social punishment institution and stretch their relative advantage (figure 2). As a consequence, more players tend to abstain from the public goods game and—more importantly—fewer players cooperate. Thus, anti-social institution supporters temporarily invade. This dynamics triggers a sizeable reduction in cooperation as shown in figure 2.

Figure 2 also shows two unstable fixed points: between the pro-social and the anti-social institution (*WCCCC* and *MDDDD*), and between the pro-social institution and defectors (*WCCCC* and *NDDDD*). The existence of these unstable fixed points can be illustrated from the competition between the two associated strategies. In the case of *WCCCC* and *NDDDD*, the associated pay-offs in a population with j cooperating players are

$$\pi_{WCCCC} = -\gamma - c + \sum_{i=0}^{n-1} \frac{\binom{j-1}{i}\binom{N-j}{n-i-1}}{\binom{N-1}{n-1}} \frac{cri}{n-1} = -\gamma - c + cr\frac{j-1}{N-1} \tag{3.1}$$

and

$$\pi_{NDDDD} = +\sum_{i=0}^{n-1} \frac{\binom{j}{i}\binom{N-j-1}{n-i-1}}{(N-1n-1)} \left(\frac{cri}{n-1} - \beta i\right) = (cr - \beta(n-1))\frac{j}{N-1}. \tag{3.2}$$

For $j = 1$, we have $\pi_{WCCCC} < \pi_{NDDDD}$ when the costs of cooperation and supporting the institution outweigh the fine imposed on the defectors and the additional benefit they get from the public goods (due to the setting where a cooperating player does not benefit from her own contribution), $-\gamma - c < (cr - \beta(n-1)/N-1)$. This condition will always be fulfilled for large N. In this case, defectors cannot be invaded by cooperating supporters of a pro-social institution. For $j = n-1$, the condition for $\pi_{WCCCC} > \pi_{NDDDD}$ reduces to $+\gamma + c < \beta(n-1)$, i.e. the costs of cooperation and supporting the institution must be smaller than the fine imposed on the defectors. Thus, cooperating supporters of a pro-social institution cannot be invaded by defectors. Since neither of the two strategies can invade the others (and since the pay-offs are linear in j), this results in a bi-stability.

A similar argument holds for the pair *MDDDD* and *WCCCC*. A more comprehensive (numerical) analysis that includes all pairs of these nine strategies is presented in the electronic supplementary material.

The paths via such bi-stabilities are not prevalent in the computer simulations that will typically follow the paths highlighted in the figure. Four different cycles are prevalent: the first one, $L \to NCCCC \to NDDDD \to L$, has already been described in Hauert *et al.* [31]; the second one, $L \to WCCCC \to NCCCC \to NDDDD \to L$, additionally emerges in the institutional punishment model of Sigmund *et al.* [6]; while the remaining two, $L \to NCCCC \to MDDDD \to L$ and $L \to NCCCC \to MDDDD \to NDDDD \to L$, emerge only in the presence of anti-social punishment institutions.

One may argue that an anti-social institution should never arise because cooperators can be invaded by defectors through an easier path without any anti-social punishment. However, this argument also applies to the transition from loners to cooperators, which could occur with a pro-social

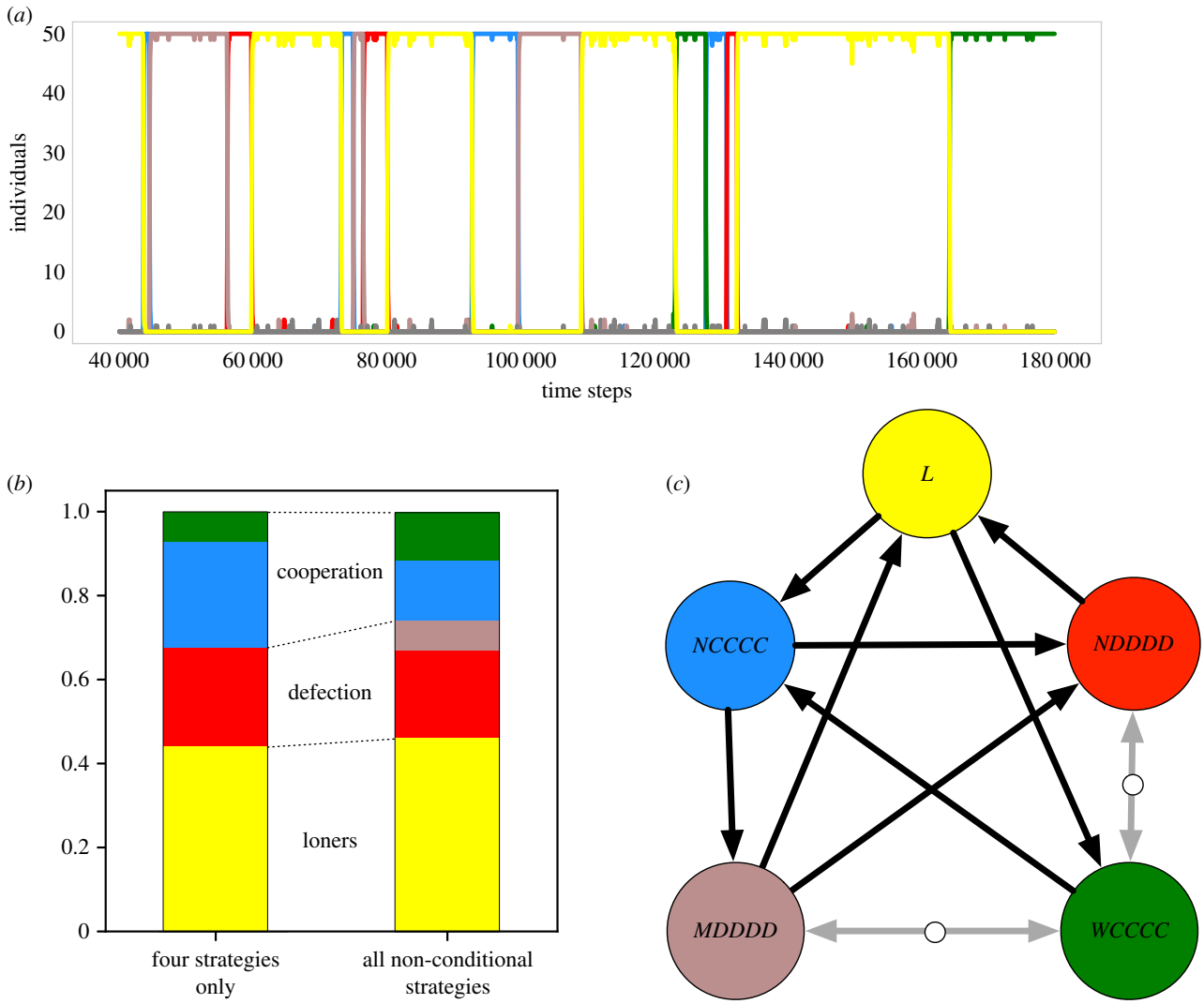

Figure 2. Evolutionary dynamics for the nine unconditional strategies. (*a*) Typical simulation run over 140 000 time steps shows different cycles in which strategies replace each other, the dominated strategies that punish themselves are only present in low abundance (colour code for the strategies as in the other panels). Because of the low mutation rate, the abundances are typically close to 0% or 100%. (*b*) The stationary distribution obtained from computer simulations—four strategies from [6] only, without anti-social punishment, versus all unconditional strategies. Averages are taken over 300 independent repetitions, each running for 5×10^6 generations, averaging the second half of each replicate. The inclusion of the anti-social institution (*MDDDD*), which from the outside is paradoxical and should play no role in evolution, reduces the level of cooperation. (*c*) Pairwise invasion diagram for the five strategies that are not dominated. Circles represent the strategies, arrows indicate the direction of selection. Bold arrows represent the paths that are prevalent in computer simulations. The dynamics can follow several intertwined cycles, e.g. $L \rightarrow NCCCC \rightarrow MDDDD \rightarrow NDDDD \rightarrow L$; see main text (we use our default set of game parameters $n = 5$, $\sigma = 1$, $c = 1$, $r = 3$, $\gamma = 0.7$, $\beta = 1.5$, population size $N = 50$, mutation rate $\mu = 0.001$, intensity of selection $\omega = 10$). (Online version in colour.)

pool, but also without any punishment. The crucial difference is that, while the pro-social pools can rise to high abundance [6], the anti-social behaviour only plays an important role in facilitating the emergence of other strategies without becoming prevalent itself. The complexity of implementing an anti-social and a pro-social institution per se is the same. Their asymmetry arises only from evolutionary competition.

Notably, figure 2*b* shows that artificially taking out seemingly unimportant strategies has an effect on the predicted level of cooperation. For our default set of parameters, we observe a slight increase in loners, from 44% to 46%. Likewise, defection increases by about 4% and the overall level of cooperation decreases by 6% through the introduction of the anti-social punishment institution.

3.2. Observable institutions

In many cases, information on punishment pools may be available before players need to make a decision on their contribution, such that players can condition their actions on the existence of institutions [26,28,29]. For example, criminals are arguably less likely to offend if they know an institution is in place to punish them [36]—although see also [37]. First, we focus on consistent conditional strategies (figure 1) which do not punish themselves: if individuals support a pro-social institution, they cooperate if that institution is in place. If individuals support an anti-social institution, they defect if that institution is in place. Thus, they do not support both institutions at the same time. Moreover, if they cooperate (defect) under a single institution, they also cooperate (defect) when both institutions exist. This set contains four strategies that support the pro-social institution, $W \circ C \circ C$ (where the entries $\circ$ are either C or D), and four strategies that support the anti-social institution, $M \circ\circ DD$. In addition, we have 12 strategies that do not support any institution, $N \circ\circ\circ\circ$, where the four strategies $N \circ CCD$ and $N \circ DDC$ are excluded. Finally, we have the option to abstain from the public goods game, L.

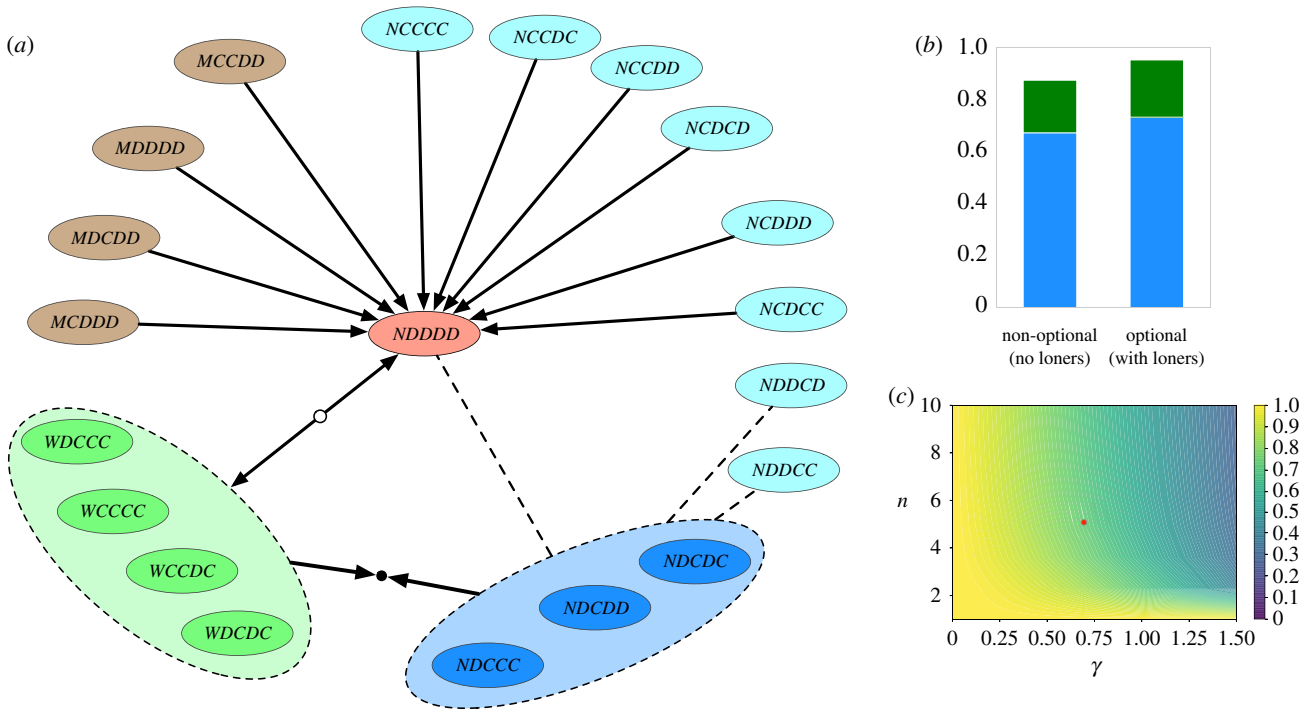

Figure 3. Evolutionary dynamics for the 20 consistent conditional strategies. (*a*) Evolutionary dynamics between pairs of strategies. We only depict invasions that are prevalent in the dynamics under strong selection. Given sufficient time, stable coexistences between two kinds of players emerge: those that support pro-social punishment and those not supporting any institution, but cooperating in the presence of pro-social punishment. For simplicity, the diagram focuses on a non-optional game: adding the loner strategy creates a fast path from defection, via loners, into either side of the stable mixture. (*b*) There are 12 different stable coexistences with similar abundance based on the four strategies supporting pro-social punishment and the three opportunistic strategies that cooperate in the presence, but defect in the absence, of coordinated pro-social punishment. We group these strategies in the cases of non-optional and optional public goods games. Averages taken as described in figure 2. (*c*) The probability that a pro-social pool is implemented in the coexistence decreases with increasing costs γ and increasing group size n. The dot indicates our default set of parameters (see main text or figure 2). (Online version in colour.)

Electronic supplementary material, figure S1 summarizes the dynamics between the associated 210 (=(1/2) × 21 × 20) pairs of consistent conditional strategies. In this set of strategies, no strategy is strictly dominated. The evolutionary dynamics is governed by stable coexistences between a minority of players that support the pro-social institution, $W \circ C \circ C$ (called I below), and opportunists that cooperate only when the pro-social institution is in place [28], $NDC \circ\circ$ (called O below). As long as a single supporter of the institution can induce its existence, $k = 1$, their pay-off is $\pi_I = r\ c - c - \gamma$. The probability that a focal opportunist is in a group that contains at least one supporter of the pro-social institution is $1 - x^{n-1}$, where x is the fraction of opportunists, who obtain a pay-off $rc - c$ in that case. If opportunists are alone, no one cooperates and their pay-off is zero, such that their average pay-off becomes $\pi_O = (rc - c)(1 - x^{n-1})$. The condition $\pi_I = \pi_O$ leads to a unique stable equilibrium,

$$x^* = \left(\frac{\gamma}{rc - c}\right)^{1/(n-1)}. \tag{3.3}$$

The probability that an institution is implemented in a group is then $1 - (x^*)^n$, i.e. a small fraction of supporters of the institution can induce high levels of cooperation (figure 3).

The resulting stable coexistences are remarkably resilient to evolutionary invasions. We can see this by doing a pairwise analysis, and considering strategies $W \circ C \circ C$ and $NDC \circ\circ$, where the entry in $\circ$ is irrelevant (and the last entry in $W \circ C \circ C$ follows from our restriction to consistent strategies). This coexistence cannot be invaded by any other single mutant.

— Players that cooperate in the absence of an institution, $NCC \circ\circ$, would be exploited by the $NDC \circ\circ$ resident.
— Those that defect in the presence of an institution, $NDD \circ\circ$, would suffer from punishment.
— Any player supporting the anti-social institution would obtain a lower pay-off than the players in the stable coexistence: there, both players obtain $-\gamma + rc - c$. A single supporter of the anti-social institution would at most get $-\gamma + r\ c - \beta$. This assumes that, in the presence of both institutions, all the other players cooperate and that at least one individual contributed to a pro-social institution. As $\beta > c$, this pay-off is always smaller than the pay-off of the two resident types. Therefore, supporters of an anti-social institution cannot invade.

Thus, the coexistence between $W \circ C \circ C$ and $NDC \circ\circ$ is stable against single mutants. As we have made no assumption on $\circ$, this holds for all such strategies. There is no other pairwise stable coexistence in the system.

Although this theoretical analysis assumes that populations are very large, our simulation results for $N = 50$ perfectly match the prediction (figure 3*b*). This is due to the fact that selection is strong in the simulations. As a general rule of thumb, we expect this prediction to hold whenever the product of intensity of selection and population size is large [38]. This relationship between infinite and finite populations has been studied in detail elsewhere; see [39].

Note that, with conditional strategies, defection can be left via neutral paths towards strategies that do not support any institution and do not cooperate in the absence of institutions—but potentially in their presence. This implies that

the game no longer needs to be optional for cooperation to evolve [28], a potential issue with previous models [40]. Figure 3 shows the strategies in this set of 20 strategies as well the typical evolutionary dynamics between pairs of strategies.

Our model assumes that, if it is costly to make institutions visible, the cost is part of the funds paid in order to establish the institution. However, it is also possible to assume that this cost is paid by strategies using conditional information. This extension is discussed in the electronic supplementary material.

3.3. Including all conditional strategies

So far, we have given arguments that allow us to focus on specific subsets of strategies. To verify the robustness of our findings, we also consider the full strategy set of 65 strategies and implement the same computational evolutionary model as before. Because this strategy space includes many new possibilities, a plethora of different combinations could evolve. However, the same coexistence between individuals supporting pro-social institutions and opportunists not supporting any institution is found again, with the same 12 stable coexistences as in the subset of the 20 consistent strategies considered above. Thus, the evolutionary outcome is independent of the choice of the strategy subset in our model once the key opportunistic strategies are considered. Figure 4 shows a typical time series of the dynamics with a complete strategy space. The system spends the vast majority of time in stable coexistences, where the strategies are occasionally replaced by others that display the same behaviour in this situation.

3.4. Higher thresholds for punishment implementation

When a certain number of supporters is required to implement an institution, all players can find themselves in groups with no punishment in place. This implies that the behaviour in such situations is under selection. Qualitatively, the results remain identical to the case where punishment can be implemented by a single supporter, $k = 1$, but the position of the fixed point and the level of cooperation can change. The structure of the game resembles a threshold public goods game [41,42]; see electronic supplementary material. When at least two supporters of an institution are needed, $k = 2$, the equilibrium fraction of players supporting the pro-social set-up increases from $\approx 23\%$ ($k = 1$) to $\approx 43\%$ ($k = 2$). At the same time, the probability that the pool is actually implemented decreases slightly from $\approx 73\%$ ($k = 1$) to $\approx 72\%$. A pairwise analysis of strategies reveals that now other coexistences are possible as well, for example between *WCCDD* and *NDCDD* (see figure 1 for an explanation of the strategy notation). In the electronic supplementary material, we show that, for $k = 2$, all these additional coexistences are unstable with respect to the invasion of a third strategy. Only the coexistences discussed above for the case of $k = 1$ remain stable against all invasions.

The equilibrium that sustains cooperation resembles that arising in a volunteer's dilemma [43], where the volunteering threshold, k, represents the number of contributors required to establish a punishment institution. We also note that the possibility of these types of coexistences has been discussed in the context of nonlinear public good games [44]. Here, we also show that this type of coexistence is particularly resilient and

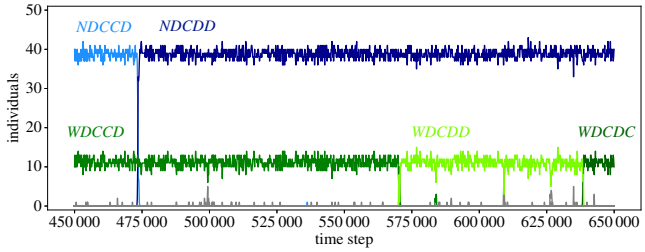

Figure 4. Time series showing a snapshot of the evolution in the complete strategy space of 65 strategies. Different instances of stable coexistences in the form *WDC* ∘ *C* and *NDC* ∘∘ are following each other, where each strategy can be replaced independently of the other. For instance, here *NDCCD* is replaced by *NDCDD* before replacements within the institutional supporters take place (from *WDCCD* via *WDCDD* to *WDCDC*). For the present parameter set (see main text), 23% of institutional supporters induce cooperation in 73% of all games (window over 2×10^5 time steps, $k = 1$). (Online version in colour.)

can arise with many flavours in the context of pool punishment, i.e. different combinations of conditional strategies that can resist invasions arising from a large strategy space. Our simulations also show that this equilibrium is stable when the population is finite and includes demographic noise.

4. Discussion

We find that pro-social, but not anti-social, punishment emerges based on individual-level selection with full symmetry between the two kinds of punishment institutions. Evolutionary dynamics introduces a symmetry breaking between the two kinds of strategies selected that favours pro-social states. The prevalent outcome is a stable coexistence between cooperators supporting a pro-social pool and those willing to cooperate when such institutions are in place, but not cooperating otherwise. For peer punishment, such stable coexistences between strategies do not appear, because they require an honest signal prior to the game [45,46].

Typical models in evolutionary game theory restrict themselves to a small number of strategies that seem to be interesting from the outset, which greatly facilitates the analysis [47]. While such a restriction can be highly insightful, the conclusions from the model can in some cases strongly depend on the strategy set [23,24,48–50]. Our computational model implements the entire possible set of strategies in the context of coordinated, institutional punishment. One may be tempted to exclude strategies that seem illogical in the context of the model, but this can be misguided. Selecting a behaviour necessarily implies that other behaviours are driven out. Therefore, the absence of behaviour should be the result of evolutionary competition and not a result of the modeller's subjective choice. As shown in this paper, strategies can have a strong impact on the evolutionary outcome even if they do not rise to high average abundances and only temporarily pave the way for other strategies. We believe this robustness test is important in simple models, which may otherwise have biased conclusions.

In our model, the maintenance of pro-social punishment relies on a small minority which supports them, but their robustness stems from the fact that they are constantly challenged by the presence of players that stop cooperating in the absence of pro-social institutions. This combination is

empirically prevalent, with resilient public goods often being supported by a minority of contributors [51]. The key to the maintenance of the public good is the observability of institutions: only public knowledge of the presence of punishment institutions allows the conditional strategies that ultimately prevent the rise of anti-social behaviour, either in the form of defection or in the form of coordinated anti-social punishment institutions that never rise to high abundance, but can undermine the public good.

An interesting problem arising here is the possible effect of institutional asymmetries. For example, anti-social institutions may entail higher costs than pro-social ones, or implement asymmetric fines in which contributors and non-contributors are punished differently. In particular, anti-social institutions may offer an evolutionary advantage if they level intrinsic asymmetries between players.

Our model is primarily concerned with the emergence and establishment of primitive institutions, thus we do not explicitly model implementation details. Instead, we assume that, once implemented, institutions for punishment will work as expected. In reality, additional issues may also arise, e.g. due to corruption [8,52,53] or group heterogeneity [54]. Asymmetries, in particular, have been shown to be important in infinite populations [55]. The tools necessary to study finite evolutionary dynamics arising from asymmetric games are not fully developed yet [56].

Most modern societies put law enforcement into the hands of institutions and do not allow their citizens to punish others directly. Our model suggests that our instinct against taking the law into our own hands is justified: the value of the signal conferred by the presence of pro-social punishment institutions may be crucial in promoting the kind of cooperation observed in humans.

Data accessibility. Our simulation code is available at https://osf.io/6tjmb/.

Authors' contributions. J.G. and A.T. developed the model together and wrote the paper together. J.G. performed numerical simulations, A.T. performed analytical approximations.

Competing interests. We declare we have no competing interests.

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Electronic Supplementary Material: Evolution of coordinated punishment to enforce cooperation from an unbiased strategy space

Julián García and Arne Traulsen

A Evolutionary Dynamics

We assume that successful strategies spread in a finite population in proportion to their relative fitness. For each one of the strategy sets, we consider a Moran process with exponential payoff to fitness mapping [1, 2]:

- We calculate the average payoff π of each individual given the current composition of the population. Fitness is then defined as $f = \exp[+\omega\pi]$
- One individual is selected at random, proportional to fitness. With probability $1 - \mu$, the offspring is identical to the parent. With probability μ, its offspring is mutated and is assigned a new random type (selected among all types, including the parents).
- The offspring replaces a randomly chosen individual.

An analytical approach based on small mutation rates [3] is problematic for the present model, as there are stable mixtures of strategies which call for mutation rates exponentially small in the population size and the intensity of selection [4]. Also a weak selection analysis for any mutation rate makes limited sense in our case [5], as this is based on deviations from a state where a large number strategies would coexist at identical abundance – a situation that appears ill-fitted to our setup with a large number of strategies. To bypass these difficulties, we focus on simulations to single out the important parts of the dynamics, which we then study analytically.

B Parameter choice

For the evolutionary dynamics, we assume a population of size $N = 50$, a mutation rate $\mu = 10^{-3}$, and a high intensity of selection $\omega = 10$ to obtain results robust under changes of the parameters of the game [6, 7]. Note that the model is interesting only if these parameters fulfill the following four conditions

(i) A group in which all individuals contribute to a pro-social pool cooperates and is better off than a group in which everyone abstains, i.e., $rc - c - \gamma > \sigma$.

(ii) It is better to abstain than to be in a group of defectors, $\sigma > 0$.

(iii) The fine exceeds the costs, $\beta > c$, i.e., punishment is always a deterrent, since the fine paid for not contributing outweighs contributions themselves.

(iv) The fine exceeds the cost of punishing someone, $\beta > \gamma$ and the cost of contribution is smaller than the cost of punishing, $\gamma < c$.

For our simulations, the game parameters are as chosen as follows: The loner payoff is $\sigma = 1$. Contribution to the public good is $c = 1$, and the multiplier is $r = 3$. The individual cost of supporting a sanctioning pool is $\gamma = 0.7$. A fine per unit of contribution to a sanctioning pool is $\beta = 1.5$. In our simulations, we work with a population size of $N = 50$ and a group size of $n = 5$.

C Analysis of pairwise coexistences for $k = 1$

C.1 Pairwise coexistences for k=1.

NCCDC	– – – –	NCCCC	NDCCC	⟵	NCCDD	WDCDC	⟵ ○ ⟶	NCDCD	WDCDC	⟶ ● ⟵	NDCDC	WDCDC	⟵ ○ ⟶	NDDDD
NCCDD	– – – –	NCCCC	NDCDC	⟵	NCCDD	MCCDD	⟵	NCDCD	MCCDD	⟶	NDCDC	MCCDD	⟶	NDDDD
NCDCC	– – – –	NCCCC	NDCDD	⟵	NCCDD	MCDDD	⟵	NCDCD	MCDDD	⟶	NDCDC	MCDDD	⟶	NDDDD
NCDCD	– – – –	NCCCC	NDDCC	⟵	NCCDD	MDCDD	⟵	NCDCD	MDCDD	⟶	NDCDC	MDCDD	⟶	NDDDD
NCDDD	– – – –	NCCCC	NDDCD	⟵	NCCDD	MDDDD	⟵	NCDCD	MDDDD	⟶	NDCDC	MDDDD	⟶	NDDDD
NDCCC	⟵	NCCCC	NDDDD	⟵	NCCDD	L	⟶	NCDCD	L	⟵	NDCDC	L	⟵	NDDDD
NDCDC	⟵	NCCCC	WCCCC	⟶	NCCDD	NDCCC	⟵	NCDDD	NDDCC	– – – –	NDCDD	WCCDC	– – – –	WCCCC
NDCDD	⟵	NCCCC	WCCDC	⟶	NCCDD	NDCDC	⟵	NCDDD	NDDCD	– – – –	NDCDD	WDCCC	– – – –	WCCCC
NDDCC	⟵	NCCCC	WDCCC	⟶	NCCDD	NDCDD	⟵	NCDDD	NDDDD	– – – –	NDCDD	WDCDC	– – – –	WCCCC
NDDCD	⟵	NCCCC	WDCDC	⟶	NCCDD	NDDCC	⟵	NCDDD	WCCCC	⟶ ● ⟵	NDCDD	MCCDD	⟵ ○ ⟶	WCCCC
NDDDD	⟵	NCCCC	MCCDD	⟶	NCCDD	NDDCD	⟵	NCDDD	WCCDC	⟶ ● ⟵	NDCDD	MCDDD	⟵ ○ ⟶	WCCCC
WCCCC	⟶	NCCCC	MCDDD	⟶	NCCDD	NDDDD	⟵	NCDDD	WDCCC	⟶ ● ⟵	NDCDD	MDCDD	⟵ ○ ⟶	WCCCC
WCCDC	⟶	NCCCC	MDCDD	⟶	NCCDD	WCCCC	⟵ ○ ⟶	NCDDD	WDCDC	⟶ ● ⟵	NDCDD	MDDDD	⟵ ○ ⟶	WCCCC
WDCCC	⟶	NCCCC	MDDDD	⟶	NCCDD	WCCDC	⟵ ○ ⟶	NCDDD	MCCDD	⟶	NDCDD	L	⟶	WCCCC
WDCDC	⟶	NCCCC	L	⟶	NCCDD	WDCCC	⟵ ○ ⟶	NCDDD	MCDDD	⟶	NDCDD	WDCCC	– – – –	WCCDC
MCCDD	⟵	NCCCC	NCDCD	– – – –	NCDCC	WDCDC	⟵ ○ ⟶	NCDDD	MDCDD	⟶	NDCDD	WDCDC	– – – –	WCCDC
MCDDD	⟵	NCCCC	NCDDD	– – – –	NCDCC	MCCDD	⟶	NCDDD	MDDDD	⟶	NDCDD	MCCDD	⟵ ○ ⟶	WCCDC
MDCDD	⟵	NCCCC	NDCCC	⟵	NCDCC	MCDDD	⟶	NCDDD	L	⟵	NDCDD	MCDDD	⟵ ○ ⟶	WCCDC
MDDDD	⟵	NCCCC	NDCDC	⟵	NCDCC	MDCDD	⟶	NCDDD	NDDCD	– – – –	NDDCC	MDCDD	⟵ ○ ⟶	WCCDC
L	⟶	NCCCC	NDCDD	⟵	NCDCC	MDDDD	⟶	NCDDD	NDDDD	– – – –	NDDCC	MDDDD	⟵ ○ ⟶	WCCDC
NCCDD	– – – –	NCCDC	NDDCC	⟵	NCDCC	L	⟶	NCDDD	WCCCC	⟵ ○ ⟶	NDDCC	L	⟶	WCCDC
NCDCC	– – – –	NCCDC	NDDCD	⟵	NCDCC	NDCDC	– – – –	NDCCC	WCCDC	⟵ ○ ⟶	NDDCC	WDCDC	– – – –	WDCCC
NCDCD	– – – –	NCCDC	NDDDD	⟵	NCDCC	NDCDD	– – – –	NDCCC	WDCCC	⟵ ○ ⟶	NDDCC	MCCDD	⟵ ○ ⟶	WDCCC
NCDDD	– – – –	NCCDC	WCCCC	⟵ ○ ⟶	NCDCC	NDDCC	– – – –	NDCCC	WDCDC	⟵ ○ ⟶	NDDCC	MCDDD	⟵ ○ ⟶	WDCCC
NDCCC	⟵	NCCDC	WCCDC	⟵ ○ ⟶	NCDCC	NDDCD	– – – –	NDCCC	MCCDD	⟵	NDDCC	MDCDD	⟵ ○ ⟶	WDCCC
NDCDC	⟵	NCCDC	WDCCC	⟵ ○ ⟶	NCDCC	NDDDD	– – – –	NDCCC	MCDDD	⟵	NDDCC	MDDDD	⟵ ○ ⟶	WDCCC
NDCDD	⟵	NCCDC	WDCDC	⟵ ○ ⟶	NCDCC	WCCCC	⟶ ● ⟵	NDCCC	MDCDD	⟵	NDDCC	L	⟶	WDCCC
NDDCC	⟵	NCCDC	MCCDD	⟵	NCDCC	WCCDC	⟶ ● ⟵	NDCCC	MDDDD	⟵	NDDCC	MCCDD	⟵ ○ ⟶	WDCDC
NDDCD	⟵	NCCDC	MCDDD	⟵	NCDCC	WDCCC	⟶ ● ⟵	NDCCC	L	⟵	NDDCC	MCDDD	⟵ ○ ⟶	WDCDC
NDDDD	⟵	NCCDC	MDCDD	⟵	NCDCC	WDCDC	⟶ ● ⟵	NDCCC	NDDDD	– – – –	NDDCD	MDCDD	⟵ ○ ⟶	WDCDC
WCCCC	⟶	NCCDC	MDDDD	⟵	NCDCC	MCCDD	⟵	NDCCC	WCCCC	⟵ ○ ⟶	NDDCD	MDDDD	⟵ ○ ⟶	WDCDC
WCCDC	⟶	NCCDC	L	⟶	NCDCC	MCDDD	⟵	NDCCC	WCCDC	⟵ ○ ⟶	NDDCD	L	⟶	WDCDC
WDCCC	⟶	NCCDC	NCDDD	– – – –	NCDCD	MDCDD	⟵	NDCCC	WDCCC	⟵ ○ ⟶	NDDCD	MCDDD	– – – –	MCCDD
WDCDC	⟶	NCCDC	NDCCC	⟵	NCDCD	MDDDD	⟵	NDCCC	WDCDC	⟵ ○ ⟶	NDDCD	MDCDD	– – – –	MCCDD
MCCDD	⟶	NCCDC	NDCDC	⟵	NCDCD	L	⟵	NDCCC	MCCDD	⟵	NDDCD	MDDDD	– – – –	MCCDD
MCDDD	⟶	NCCDC	NDCDD	⟵	NCDCD	NDCDD	– – – –	NDCDC	MCDDD	⟵	NDDCD	L	⟵	MCCDD
MDCDD	⟶	NCCDC	NDDCC	⟵	NCDCD	NDDCC	– – – –	NDCDC	MDCDD	⟵	NDDCD	MDCDD	– – – –	MCDDD
MDDDD	⟶	NCCDC	NDDCD	⟵	NCDCD	NDDCD	– – – –	NDCDC	MDDDD	⟵	NDDCD	MDDDD	– – – –	MCDDD
L	⟶	NCCDC	NDDDD	⟵	NCDCD	NDDDD	– – – –	NDCDC	L	⟵	NDDCD	L	⟵	MCDDD
NCDCC	– – – –	NCCDD	WCCCC	⟵ ○ ⟶	NCDCD	WCCCC	⟶ ● ⟵	NDCDC	WCCCC	⟵ ○ ⟶	NDDDD	MDDDD	– – – –	MDCDD
NCDCD	– – – –	NCCDD	WCCDC	⟵ ○ ⟶	NCDCD	WCCDC	⟶ ● ⟵	NDCDC	WCCDC	⟵ ○ ⟶	NDDDD	L	⟵	MDCDD
NCDDD	– – – –	NCCDD	WDCCC	⟵ ○ ⟶	NCDCD	WDCCC	⟶ ● ⟵	NDCDC	WDCCC	⟵ ○ ⟶	NDDDD	L	⟵	MDDDD

ESM Figure 1: In an infinite population, a classification between all pairs of consistent conditional strategies as neutral (– – – –), dominance (⟶ or ⟵), bistabilities (⟵ ○ ⟶), or coexistences (⟶ ● ⟵) is possible. These results do not change with the parameters, but require $k = 1$. If the population size is no longer infinite, but large compared to the group size, the classification also determines the evolutionary dynamics in finite populations via fixation probabilities and fixation times.

As discussed in the main text, when a single individual can trigger the existence of a pool, $k = 1$, all pairwise stable coexistences between opportunists and supporters of the

pro-social institution display the same behaviour. Supporters of the pro-social pool always find themselves in fully cooperative groups, leading to a payoff

$$\pi_I = rc - c - \gamma. \tag{1}$$

To simplify the analysis, we assume that the population is so large that we can assume it is infinite. Then, the probability that an opportunist faces a group with at least one supporter of the pro-social pool is $1 - x^{n-1}$, where x is the fraction of opportunists, who obtain a payoff $rc - c$ in that case. If opportunists are alone, no one cooperates and their payoff is zero. On average, this leads to a payoff

$$\pi_O = (rc - c)\left(1 - x^{n-1}\right) \tag{2}$$

In the unique stable equilibrium x^* of the associated replicator dynamics [8], both payoffs must be equal, leading to

$$x^* = \left(\frac{\gamma}{rc - c}\right)^{1/(n-1)}. \tag{3}$$

Supporters of pro-social sanctions can invade opportunists (as they immediately induce cooperation) and vice versa (as opportunists save the cost γ). For $k = 1$, the game is identical to a volunteer's dilemma [9, 10, 11].

D Pairwise coexistences for $k > 1$

Let us focus on the coexistences between supporters of the sanctioning pools and opportunists, $NDC \circ \circ$, in the case where more than one individual is needed to implement punishment, $k > 1$. We assume that only effective pools that are actually realised cause any costs.

We assume that supporters of the pool defect if the pool is not implemented successfully, i.e. we focus on strategies $WDC \circ C$ (the strategies $WCC \circ C$ can be invaded by $WDC \circ C$ for $k > 1$). We abbreviate the payoff of the $WDC \circ C$ strategies as

$$\pi_{ID} = (rc - c - \gamma) \sum_{j=k-1}^{n-1} \binom{n-1}{j} (1-x)^j x^{n-1-j}, \tag{4}$$

where the sum captures the probability that a pool is realised — otherwise their payoff is zero. The opportunists obtain

$$\pi_O = (rc - c) \sum_{j=k}^{n-1} \binom{n-1}{j} (1-x)^j x^{n-1-j}. \tag{5}$$

Equating the payoffs, $\pi_{ID} = \pi_O$, leads to an expression for the fixed point,

$$\frac{\gamma}{rc - c} = \frac{\binom{n-1}{k-1} x^{n-k} (1-x)^{k-1}}{\sum_{j=k-1}^{n-1} \binom{n-1}{j} x^{n-j-1} (1-x)^j} \tag{6}$$

The right hand side of this equation increases monotonously from 0, for $x = 0$, to 1 for $x = 1$, see Peña *et al.* [12]. Thus, the associated replicator dynamics [8] has a unique interior fixed point for $\gamma < rc - c$. This needs to be solved numerically for an arbitrary $k > 1$, but for $k = n - 1$ we reach the expression $x = \gamma/(c(n-1)(r-1) - \gamma(n-2))$. This fixed point is stable, because supporters of the pool can invade a monomorphous population of opportunists (as they sometimes benefit from inducing cooperation) and opportunists can also invade a monomorphous population of institution supporters (as they save the cost γ). The probability that a pro-social pool is successfully implemented is given by $\sum_{j=k}^{n} \binom{n}{j} x^{n-j}(1-x)^j$, where x is obtained from solving Eq 6.

D.1 Pairwise coexistences for $k = 2$ that cannot be invaded

For $k = 2$, there is again a coexistence between $WDC \circ C$ and $NDC \circ \circ$. Even for $k = 2$, this coexistence cannot be invaded by other strategies for our default parameters ($n = 5$, $r = 3$, $c = 1$, $\gamma = 0.7$, $\beta = 1.5$):

- **Invasion of other supporters of pro-social pools:** If $WDC \circ C$ players would cooperate in the absence of sanctions, they would be exploited, reducing their payoff.
- **Invasion of other players who do not support any institution:** If $NDC \circ \circ$ players would cooperate in the absence of sanctions, they would be exploited. If $NDC \circ \circ$ players would defect in the presence of a sanctioning pool, they would suffer from punishment.
- **Invasion of supporters of anti-social pools:** We assume that invaders continue to defect in the absence and cooperate in the presence of the pro-social pool, i.e. we focus on $MDCDD$. This provides the scenario most attractive for the supporters of anti-social sanctions. If a player would start to support the anti-social pool, her payoff would be identical to the $NDC \circ \circ$ players as long as no sanctions are implemented (invasion is neutral until the anti-social pool is implemented). We then compare the payoff these players can obtain to the one of the residents in their equilibrium, which implies that the number of mutants remains small.

  If the mutants find themselves in groups with only the anti-social pool, they pay the cost γ, but do not induce any cooperation and thus have a disadvantage compared to the residents. If the mutants find themselves in groups with both institutions, they are punished in addition. Since in the pairwise coexistence they invade, no player is punished, the fine implies a disadvantage for the supporters of the anti-social institution.

Thus, the coexistence between $WDC \circ C$ and $NDC \circ \circ$ is stable to any invasion also for $k = 2$.

D.2 Pairwise coexistences for $k = 2$ that can be invaded

A pairwise analysis of strategies for our default parameters ($n = 5$, $r = 3$, $c = 1$, $\gamma = 0.7$, $\beta = 1.5$) and $k = 2$ reveals that in addition to $WDC \circ C$ and $NDC \circ \circ$, other stable coexistences between two strategies are possible:

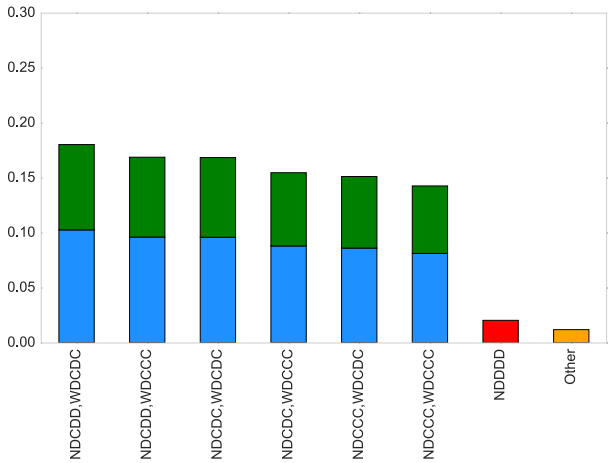
1. Coexistences between supporters of the pro-social pool and those who support no pools
   - A coexistence between $WDC \circ C$ and $NCC \circ \circ$ can be invaded by $NDC \circ \circ$ strategies.
   - A coexistence between $WDC \circ C$ and $NCD \circ \circ$ can be invaded by $NDD \circ \circ$ strategies, displaying unconditional defection.
2. Coexistences between supporters of the anti-social pool and those who support no sanctioning pools
   - A coexistence between $MD \circ DD$ and $NC \circ C\circ$ can be invaded by $ND \circ C\circ$.
   - A coexistence between $MD \circ DD$ and $NC \circ D\circ$ can be invaded by $ND \circ D\circ$.
3. Coexistences between supporters of the anti-social pool and supporters of the pro-social pool. In this case, there is always at least one institution implemented successfully, leading to three possible scenarios: Each pool alone or an implementation of both pools at the same time
   - A coexistence between $M \circ CDD$ and $W \circ CCC$ can be invaded by $W \circ CDC$.
   - A coexistence between $M \circ DDD$ and $W \circ CDC$ can be invaded by $M \circ CDD$.
   - There are two possible stable coexistences between $M \circ CDD$ and $W \circ CDC$. If the abundance of $M \circ CDD$ is high, it is more likely that the anti-social pool is implemented. Then, $N \circ CDD$ can invade. If the abundance of $W \circ CDC$ is high, it is more likely that the pro-social pool is implemented. Then, $N \circ CDC$ can invade.

   Thus, all coexistences that emerge for $k = 2$ that appear to be stable in a pairwise analysis are unstable with respect to a third invader strategy. Therefore, the qualitative results do no not change when we switch from $k = 1$ to $k = 2$, see Fig. 2. However, as there is now selection on the first bit of the pool supporter strategies to defect in the absence of sanctions, we only have six possible coexistences (displaying exactly the same behavior as for $k = 1$) instead of twelve.

E Additional costs for the sanctioning pools

Can cooperation be maintained if the supporters of the pro-social pool have an additional disadvantage by paying the cost γ for the institution even when the pool is not implemented? For $k = 1$, this situation never occurs. Only for $k > 1$, the additional costs affect the results and leads to a Volunteers Dilemma [13, 14]. In this case, the payoff of supporters of the pro-social pool is

$$\pi_{ID} = -\gamma + (rc - c) \sum_{j=k-1}^{n-1} \binom{n-1}{j} (1-x)^j x^{n-1-j}, \tag{7}$$



ESM Figure 2: **Evolutionary dynamics for the 20 consistent conditional strategies with $k = 2$.** More than 90% of the time the population is made up of a coexistence of conditional cooperators and sanctioning supporters. This indicates that the our results are robust to increasing the threshold of individuals required to establish punishment. The frequencies of cooperators and defectors when $k = 2$ indicate a similar pattern to that resulting from $k = 1$, i.e., a minority of sanctioning supporters coexists with a large number of conditional cooperators supporting no institution.

where the cost γ now always has to be payed. The payoff of opportunists remains unchanged. Now, $\pi_O = \pi_{ID}$ simplifies to

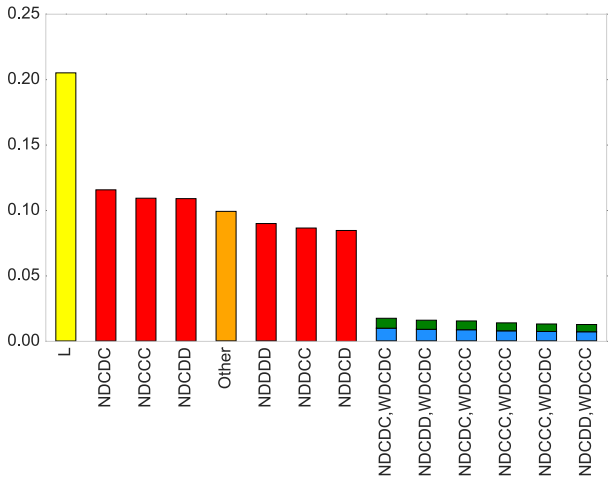
$$\frac{\gamma}{rc - c} = \binom{n-1}{k-1} x^{n-k}(1-x)^{k-1}. \tag{8}$$

The right hand side of this equation has a unique maximum at $x^* = \frac{n-k}{n-1}$ given by

$$\Gamma = \binom{n-1}{k-1} \left(\frac{k-1}{n-1}\right)^{k-1} \left(\frac{n-k}{n-1}\right)^{n-k}. \tag{9}$$

For $\Gamma > \frac{\gamma}{rc-c}$, the coexistence equilibrium vanishes and opportunists dominate over punishment supporters – no cooperation emerges. For $\Gamma < \frac{\gamma}{rc-c}$, there are two internal equilibria, one stable and one unstable. This implies that opportunists can invade and coexist with sanction supporters. But in contrast to the case of $k = 1$ and our usual assumption not implementing costs for a non-existing pool, now opportunism is stable and a substantial fraction of sanction supporters is necessary to reach the stable coexistence. This results in a complete breakdown of cooperation.

In this case — as in the case of unconditional strategies — the option not to take part in the game can provide an escape hatch out of mutual defection [15, 16]. If we add loners to our simulations, the system shows again some cooperation. Loners are necessary, as in the case of $k > 1$, opportunists can no longer invade defectors and an alternative path out of defection is necessary. However, for $k = 2$ and additional costs for the pools their success is very limited, see Fig. 3.



ESM Figure 3: **Additional costs for the institution in the case of $k = 2$.** When the pool also causes costs if its implementation fails, cooperation breaks down entirely in the absence of loners. Thus, we explore the role of loners in our simulations here and this explore the consistent conditional strategy set with 21 strategies. While loners provide a path out of defection, their success remains very limited here, as indicated by the low abundance of cooperative coexistences in this figure ($k = 2$, otherwise we use the standard parameter set).

F Additional costs on conditional strategies

Our model assumes that the cost for institutions to be visible is part of the funds that establish the institutions. However, it is also possible to assume that this cost is paid by those strategies that use conditional information. We can study this setup by considering a a cost $\delta > 0$, paid all individuals using strategies that discriminate. We will see that provided that the game is optional, the results remain valid in a wide range of parameters.

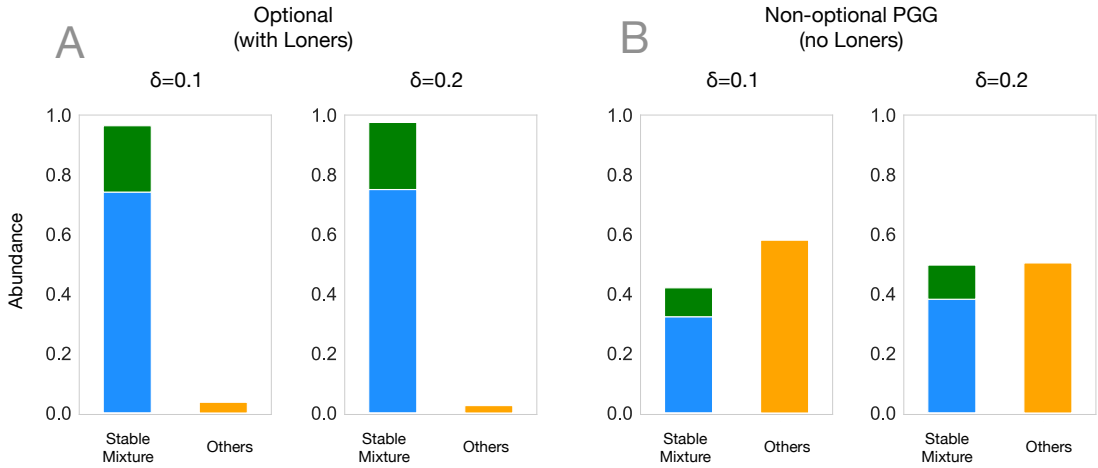
For the new cost it is important that δ is such that $0 < \sigma < (r-1)c - G - \delta$. That is, conditional cooperators in groups of cooperators still earn more than those that do not play the game.

The first thing to know is that the stable mixture that guarantees cooperation remains an equilibrium, provided $\delta < \beta - c$. First note that with the exception of $WCCCC$, any two strategies $W \circ C \circ C$ and $NDC \circ \circ$ will both pay the cost δ.

This coexistence cannot be invaded by any other single mutant:

- Individuals cooperating in the absence of an institution, $NCC \circ \circ$, would be exploited by the $NDC \circ \circ$ resident.
- Defectors in the presence of an institution, $NDD \circ \circ$, would be punished.
- Any player supporting the anti-social institution is outperformed by individuals in the stable coexistence: There, both players obtain $-\gamma + rc - c - \delta$. A single supporter of the anti-social institution would at most get $-\gamma + rc - \beta$. Therefore, supporters of an anti-social institution cannot invade provided $\delta < \beta - c$.

In addition for this equilibrium to be reached we require that the public goods game is optional. Our simulations show that the typical path into the stable mixture is via defection, which is neutral to $NDD \circ \circ$ (see Figure 3 in the main text). This path is no longer neutral if $NDD \circ \circ$ pays a cost δ, thus, getting into the mixture requires the loner strategy (see Figure 1 in this supplement).



ESM Figure 4: **Additional costs for the conditional strategies.** The stable mixture is prevalent, provided the public goods game is optional (panel A), but is less prominent when the public goods game is mandatory (panel B). We show results for $\delta = 0.1$ and $\delta = 0.2$, other parameters as before, and averaging as before.

Figure 4 shows how simulations align with this prediction. A cost in the provided range leads to qualitatively similar results provided the public goods game is optional.

G Computational methods

The simulations reported follow a standard agent-based simulation of a Moran process. Fitness inside groups is deterministic and can be obtained using an algorithm that given a group composition computes the size of the pro-social and anti-social pool as well as the number of PGG contributors.

Given the payoff inside groups we can use a multivariate hypergeometric distribution in order to estimate the probability of each possible group composition. This is efficient when the number of types is small. When the population is composed of three or more types, we resort to Montecarlo simulations to estimate group sampling probabilities out of 1000 samples. However, the payoff computation is exact for the majority of simulation runs, since a small mutation rate guarantees that the population is at most dimorphic for most of the time.

The process that estimates the frequency of mixtures works in the following fashion. A file reporting a complete simulation run is post-processed as follows. At each reported time step, we remove strategies that are below a given noise threshold and compute the support of the population without noise. We then count how many times each support

appears as a fraction of all the time-steps in the simulation. Supports of size one correspond to monomorphous populations. Each support of size two corresponds to a potential stable mixture between two strategies. Supports of size three are reported, but their abundance is negligible given that mutation rates are small. The code from the simulations is available online. Payoff inside groups is determined with the following algorithm.

```
def payoff_against_group(focal_index, group_of_others, list_of_strategies):

    # create a copy of the group including the focal individual
    group_composition = copy(group_of_others)
    group_composition[focal_index] += 1

    # is the focal player a loner
    if list_of_strategies[focal_index] == 'L':
        return sigma

    # compute effective group size
    effective_group_size = len(group_composition) - number_of_loners
    if effective_group_size <= 1:
        # there is no game
        return sigma

    # determine the sizes of pools
    prosocial_pool_size = 0
    antisocial_pool_size = 0
    is_focal_a_prosocial_supporter = False
    is_focal_an_antisocial_supporter = False

    for i in range(0, len(group_composition)):
        if group_composition[i] > 0:
            if is_i_prosocial_supporter(i):
                prosocial_pool_size = prosocial_pool_size + group_composition[i]
                if i == focal_index:
                    is_focal_a_prosocial_supporter = True
            if is_i_antisocial_supporter(i):
                antisocial_pool_size = antisocial_pool_size + group_composition[i]
                if i == focal_index:
                    is_focal_an_antisocial_supporter = True

    # check if the threshold was reached
    if prosocial_pool_size < k:
        prosocial_pool_size = 0
    if antisocial_pool_size < k:
        antisocial_pool_size = 0

    # determine size of the PGG
    pg_pool_size = 0
    focal_cooperated = False
    for i in range(0, len(group_composition)):
        if group_composition[i] > 0:
            if do_i_cooperate_in_group(i, prosocial_pool_size, antisocial_pool_size):
                pg_pool_size = pg_pool_size + group_composition[i]
                if focal_index == i:
                    focal_cooperated = True

    # Given pool sizes and PGG size compute costs and benefits

    #Compute benefit
    if focal_cooperated:
        benefit = r * c * ((pg_pool_size - 1) / (effective_group_size - 1))
    else:
        benefit = r * c * (pg_pool_size / (effective_group_size - 1))

    # Compute fines
    fines = 0.0
    if not focal_cooperated:
        # prosocial institution punishes
        fines = fines +  beta * prosocial_pool_size
    else:
        # antisocial institution punishes
        fines = fines + beta * antisocial_pool_size

    # Compute insitutional cost
    institutional_cost = 0
    if is_focal_a_prosocial_supporter and prosocial_pool_size != 0:
        institutional_cost = institutional_cost + gamma
    if is_focal_an_antisocial_supporter and antisocial_pool_size != 0:
            institutional_cost =  institutional_cost + gamma
    # cost of cooperation
    if focal_cooperated:
        return benefit - fines - institutional_cost -  c
    else:
        return benefit - fines - institutional_cost
```

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